بسمه تعالی

Restricted Isometry Property (RIP(s))

Def. Matrix $A_{n\times d}$ satisfies RIP(s) if for any $S \subseteq \{1, \dots, d\}$, $|S| \leqslant s$, we have

$$|||\, A_S^T A_S/n - I \,|||_2 \leqslant \delta_s .$$

Def:
$$|||M|||_2 := \max_{\|v\|=1} \underbrace{\|Mv\|_2} \qquad \sqrt{\|Mv\|_2^2} = \sqrt{v^T M^T M v}$$

M real Symmetric

$$:= \max_{\|v\|=1} |\underbrace{v^T M v}|$$

$$g(v) = v^T M v - \lambda(\|v\|^2 - 1)$$

$$\nabla_v g = 2Mv - 2\lambda v = 0 \Rightarrow Mv = \lambda v$$

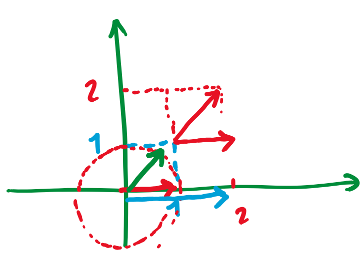
if v is eigen vec. $\Rightarrow v^T M v = v^T(\lambda v) = \lambda \|v\|_2^2 = \lambda$

$\|v\| = 1$

For the 1st opt.
$$\max_{\|v\|=1} \sqrt{v^T M^2 v}$$

$$\equiv \sqrt{\max_{\|v\|=1} v^T M^2 v} = \sqrt{\lambda_*^2} = |\lambda_*|$$

Def.. 1



$$M = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$\sqrt{2 + 1/2} = \sqrt{5/2} < 2$$

$$\max_{\|v\|=1} \sqrt{\|Mv\|_2^2} \qquad \max_{\left\|\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right\|_2 = 1} \underbrace{\left\|\begin{bmatrix} 2v_1 \\ v_2 \end{bmatrix}\right\|_2^2} \to 4v_1^2 + v_2^2 - \lambda(v_1^2 + v_2^2 - 1)$$

$$\|$$

$$\max\{2, 1\}$$

$$\|$$

$$2$$

$$8v_1 - 2\lambda v_1 = 0$$

$$2v_2 - 2\lambda v_2 = 0$$

$$\lambda = 4, 1$$

$\square$

$$RIP(s) \equiv (1-\delta_s)\|\theta\|_2^2 \leq \left\|\frac{A_S}{\sqrt{n}}\theta\right\|_2^2 \leq (1+\delta_s)\|\theta\|_2^2 \quad (*)$$

$$\forall S \subseteq \{1, \dots, d\}$$
$$|S| \leq s$$

Why these two Def. of RIP are equivalent?

subtract $\|\theta\|_2^2$ from all 3 sides of $(*)$

$$(*) \iff -\delta_s\|\theta\|_2^2 \leq \left\|\frac{A_S}{\sqrt{n}}\theta\right\|_2^2 - \|\theta\|_2^2 \leq \delta_s\|\theta\|_2^2$$

$$\iff \left| \underbrace{\left\|\frac{A_S}{\sqrt{n}}\theta\right\|_2^2}_{\frac{\theta^T A_S^T A_S \theta}{n}} - \|\theta\|_2^2 \right| \leq \delta_s \cancel{\|\theta\|_2^2}$$

$$\underbrace{\frac{\theta^T A_S^T A_S \theta}{n} - \theta^T\theta}$$

$$\underbrace{\frac{\theta^T}{\|\theta\|}}_{v^T} \Bigg[\underbrace{\frac{A_S^T A_S}{n} - I}_{M} \Bigg] \underbrace{\frac{\theta}{\|\theta\|}}_{v}$$

$$|v^T M v| \leq \delta_s$$

$$A = {}_{n}\overset{d}{\left[A_1 \,\middle|\, A_2 \,\middle|\, \cdots \qquad \,\middle|\, A_d \right]}$$

$$\|A_i\|_2^2 = n \qquad\qquad \delta_s > 0$$

RIP(1) $\quad \left\| \underbrace{\overbrace{A_i^T A_i}^{n} / n}_{1} - 1 \right\|_2 = 0 < \delta_s$

RIP(2) $\quad \left\| \underbrace{A_{(i,j)}^T A_{(i,j)} / n} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\|_2$

$$A_{(i,j)} = \left[A_i \,\middle|\, A_j \right]$$

$$\begin{bmatrix} 1 & \frac{\langle A_i, A_j \rangle}{n} \\ \frac{\langle A_i, A_j \rangle}{n} & 1 \end{bmatrix}$$

$$\left\| \underbrace{\begin{bmatrix} 0 & \langle A_i, A_j \rangle / n \\ \langle A_i, A_j \rangle / n & 0 \end{bmatrix}}_{\lambda = \pm \langle A_i, A_j \rangle / n} \right\|_2 = \left| \frac{\langle A_i, A_j \rangle}{n} \right|$$

$$\left| u^T \left(\frac{\left[A_i \,\middle|\, A_j \right]^T \left[A_i \,\middle|\, A_j \right]}{n} - I \right) \underline{u} \right|$$

$$- \|u\|_2^2$$

Theorem : If A satisfies RIP($2s$) $\delta_{2s} \leq 1/3$ then A would also satisfy NSP(S). $|S| = s$

Proof: Start with an arbitrary member of null(A), call it θ.

$$\underline{\theta} = \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_d \end{bmatrix} = \begin{array}{l} \underline{S}\{ \\ S_1\{ \\ \\ S_k\{ \end{array} \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_s \\ \hline \theta_{s+1} \\ \vdots \\ \theta_{2s} \\ \hline \vdots \end{bmatrix}$$

s elements of θ_i's with largest abs. val.

next largest

if $\rightarrow \|\underline{\theta_S}\|_1 < \|\underline{\theta_{S^c}}\|_1$ $\forall S'$ then NSP(S')

RIP$(2s)$: $(1-\delta_{2s})\|\theta_S\|_2^2 \leq \left\|\frac{A_S}{\sqrt{n}}\theta_S\right\|_2^2$

$$\Rightarrow \quad (**) \quad \|\theta_S\|_2^2 \leq \left(\frac{1}{1-\delta_{2s}}\right)\underbrace{\|A_S/\sqrt{n}\,\theta_S\|_2^2}_{\frac{1}{n}\theta_S^T A_S^T \underline{A_S\theta_S}}$$

$$\theta = \tilde{\theta}_S + \tilde{\theta}_{S_1} + \cdots + \tilde{\theta}_{S_k}$$

$$Mu = \begin{bmatrix} M_1 \cdots M_k \end{bmatrix}\begin{bmatrix} u_1 \\ \vdots \\ u_k \end{bmatrix} = u_1M_1 + \cdots + u_kM_k$$

$$\theta \in \text{null}(A) \Rightarrow A\theta = \underline{0} \quad \Rightarrow \quad A\tilde{\theta}_S = \underline{A_S\theta_S} = -\sum_{j=1}^{k} A\tilde{\theta}_{S_j}$$

$$\frac{1}{n}\theta_S^T A_S^T \underbrace{A_S\theta_S} = \frac{1}{n}\theta_S^T A_S^T\left(-\sum_{j=1}^{k} A\underbrace{\tilde{\theta}_{S_j}}_{A_{S_j}\theta_{S_j}}\right)$$

$S' = S \cup S_j$

jth element

$$\frac{1}{n}\theta_S^{aT} A_{S'}^T A_{S'}\theta_{S_j}^a$$

$$= \theta_S^{aT}\left[\frac{A_{S'}^T A_{S'}}{n} - I\right]\theta_{S_j}^a$$

$$\leq \left| \theta_s^{a^T} \left[\frac{A_{s'}^T A_{s'}}{n} - I \right] \theta_{s_j}^a \right| \leq \|\theta_s^a\|.$$

Using $(**)$: $\|\theta_s\|_2^{\cancel{2}} \leq \left(\frac{1}{1-\delta_{2s}}\right) \sum_{j=1}^{K} \cancel{\|\theta_s\|_2}\, \delta_{2s} \|\theta_{s_j}\|_2 \quad \left\| \left(\frac{A_{s'}^T A_{s'}}{n} - I \right) \theta_{s_j}^a \right\|$

Wrong Sol.

$$\frac{1}{\sqrt{s}} \|\theta_s\|_1 \leq \|\theta_s\|_2 \leq \left(\frac{1}{1-\delta_{2s}}\right) \sum \delta_{2s} \|\theta_{s_j}\|_2 \leq \frac{\delta_{2s}}{1-\delta_{2s}} \sum \|\theta_{s_j}\|_1$$

$$\underline{\underline{\leq}}\ \delta_{2s} \|\theta_{s_j}^a\|_2$$

upper bound RIP indeed!

$$\|\theta_s\|_1 \leq \sqrt{s} \frac{\delta_{2s}}{1-\delta_{2s}} \|\theta_{s^c}\|_1$$

$$\leq \frac{1}{\sqrt{s}} \|\theta_{s_{j-1}}\|_1$$

$$\leq \delta_{2s} \cancel{\|\theta_s\|_2} \|\theta_{s_j}\|_2$$

right Sol.

$$\frac{\|\theta_{s_j}\|_1}{s} \geq \|\theta_{s_{j+1}}\|_\infty$$

$$\|\theta_{s_{j+1}}\|_2 \leq \sqrt{s} \|\theta_{s_{j+1}}\|_\infty \leq \frac{\sqrt{s}}{s} \|\theta_{s_j}\|_1$$

$$\frac{\sqrt{s}}{s} \to \frac{1}{\sqrt{s}}$$

$$\checkmark \quad \leq \delta_{2s} \cancel{\|\theta_s\|_2} \overbrace{\sum_j \|\theta_{s_j}\|_2} \quad \Rightarrow \quad \frac{1}{\cancel{\sqrt{s}}} \|\theta_s\|_1 \leq \frac{\delta_{2s}}{1-\delta_{2s}(\cancel{\sqrt{s}})} \|\theta\|_1$$

$$\frac{1}{s} \|\theta_{s_j}\|_1 \geq \|\theta_{s_{j+1}}\|_\infty$$

$$\|\theta_{s_{j+1}}\|_2 \leq \sqrt{s} \|\theta_{s_{j+1}}\|_\infty \leq \underbrace{\frac{\sqrt{s}}{s}}_{\frac{1}{\sqrt{s}}} \|\theta_{s_j}\|_1$$

$$\delta_{2s} < 1/3 \quad \Rightarrow \quad \|\theta_s\|_1 < \|\theta_{s^c}\|_1$$

$\cancel{\frac{1}{\sqrt{s}}}$ LHS

RHS $(**)$

$$\sum_{j=1}^{K} \|\theta_{s_j}\|_2 \leq \cancel{\frac{1}{\sqrt{s}}} \underbrace{\sum_{j=0}^{K-1} \|\theta_{s_j}\|_1}_{\leq \underbrace{\|\theta\|_1}_{\|\theta_s\|_1 + \|\theta_{s^c}\|_1}}$$

$$\Rightarrow \quad \|\theta_s\|_1 \leq \underbrace{\frac{\delta_{2s}}{1-\delta_{2s}}}_{\leq 1/2} \left(\|\theta_s\|_1 + \|\theta_{s^c}\|_1 \right) \leq 1/2 \|\theta_s\|_1 + 1/2 \|\theta_{s^c}\|_1$$

if $\delta_{2s} < 1/3$

$$\Rightarrow \quad \cancel{1/2} \|\theta_s\|_1 \leq \cancel{1/2} \|\theta_{s^c}\|_1$$

$\square$